

TABLE I
Active Components Market (US\$ million)

Type of components	Total market	Indigenous	Imports	Exports
ICs	1163	47	1116	—
CRTs	789	631	158	7
Diodes	104	20	84	2
Transistors	83	17	66	3
LEDs	60	5	55	—
Total	2199	720	1479	12

Source: ELCINA's ELCOMOS report

TABLE II
An Overview of Indian Diode Market

Heads	Description
Overall Indian market size in FY 2009-10	\$104 million
Ratio of imports : Indian Manufacturing in FY 2009-10	80 : 20
Growth in FY 2009-10	6 per cent
Export in FY 2009-10	\$2 million
Market size by type of diode	Rectifier diodes – \$70 million Zener diodes – \$21 million Other diodes – \$13 million
Key application segments	Consumer durable and lighting
Estimated growth in FY 2010-11	16 per cent
Estimated ratio of imports : Indian manufacturing in FY 2010-11	79 : 21

Source: ELCINA's ELCOMOS report

TABLE III
Diodes Market Size by Industry Sectors and Future Growth Trends

Application sector	Diodes market size	Growth rate
Telecommunication	10 per cent	30 per cent
Automotive	5 per cent	25 per cent
Consumer durables	30 per cent	18 per cent
IT and office automation	15 per cent	20 per cent
Medical electronics	15 per cent for strategic, medical, industrial and other application sectors combined	17 per cent
Lighting		12 per cent
Industrial electronics		7 per cent
Strategic electronics		15 per cent

Source: ELCINA's ELCOMOS report

facilities are installed, promote front-end manufacturing in India."

Diodes market

According to ELCOMOS report published by ELCINA, the overall market

petitive pricing offered by products imported from China and Taiwan.

Few Indian suppliers are in the business of export, but some multinational companies with manufacturing locations in India are actively involved

size for the diodes in India was about \$104 million in FY 2009-10 with indigenous manufacturing accounting for 20 per cent of the total market. Rest of the demand is met through imports from China and Taiwan, where diodes are manufactured more competitively because of larger scale of production. It is estimated that there are close to 30 diode manufacturing units operating in India with more than 50 per cent of them being classified under small-scale industry.

The Indian diode market is estimated to have reached a value of \$140 million in FY 2011-12. Consumer durables (30 per cent) and lighting (25 per cent) are the two largest sectors that account for the major portion of the demand for diodes. Apart from consumer durables and lighting, the growth is driven by automotive and telecommunication sectors.

The demand for zener and other diodes is largely met through imports. This is largely attributed to the non-availability of right technology for production and com-

in exports from India to rest of the world.

Talking about advantages and disadvantages of diodes manufacturing in India, Atnoorkar says, "As far as power diodes are concerned, the wafer processing, assembly and test operations are labour-intensive and the volumes do not support efforts of automation of wafer processing assembly and test. So India still has some advantage and is generally in a position to take lead. However, in general, semiconductor manufacturing did not grow in India and local manufacturers do face problems like sourcing of quality raw material and qualified suppliers for all the spectrum of diodes and transistors, in general semiconductors."

In diodes, copper is used in the form of touch points. Unfortunately, India is not a large producer of copper. This is due to non-availability of copper ore in India. Copper production in India depends on the import of copper ore from countries like Canada, Indonesia, Chile and Australia.

Lack of access to cost-effective technology and machinery to produce chip-based diodes and zener diodes is the reason attributed by the Indian manufacturers for a lower contribution to indigenous production.

According to industry experts, Europe, Asia-Pacific and Japan dominate the world diode market with a combined market share of over 76 per cent. Among this, Asia is the largest regional market contributing around 60 per cent of the total demand, followed by Europe and Japan. The Indian export was around \$2 million in FY2009-10, which accounted for about 0.1 per cent of the total Asian outputs.

Trends in diodes market

With the major end-user industries focused on miniaturisation, lower power consumption, higher integration and compact packaging, the market for diodes is also expected to undergo a major shift away from leaded diodes to chip-based diodes. This has driven some of the leading players (like Continental Devices India Limited (CDIL)